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Microprocessors And Microcontrollers [BECE204L] CAT-1 C2 2023-2024

Q. No.	Questions	Max Marks
1	a) Compare i3, i5 and i7 processors in terms of cores, frequency, multi-threading, turbo boost, etc.	5
	b) Identify and rectify (if any) the errors of the following instructions ADD B, R5 SETB P1 MOVC PC, #0523 PUSH R3 MOVC @A+PC, A	5
2	Calculate the amount of delay caused by the below ALP program. Assume the crystal frequency is given as 11.059 MHz and the number of machine cycles are provided in parentheses. DELAY: MOV R0, #10100111B (1) AGAIN: MOV R7, #19H (1) HERE: MOV R2, #02 (1) BACK: NOP (1) NOP (1) DINZ R2, BACK (2) DINZ R7, HERE (2) DINZ R0, AGAIN (2) RET (2)	10
3	Write an 8051 ALP to add ten numbers saved in the code memory from the location 200H onwards and store the result in R6 and R5. Where R5 holds the lower 8-bits of the result.	10
4	Write an 8051 program to get the length (8-bit number) and breadth (8-bit number) of a rectangle from port P0 and P1 respectively. Calculate area and perimeter of the rectangle and send them through port P2 and P3 respectively. Note: assume values of the inputs are such that the outputs do not exceed 8-bits	10
5	Write an ALP for generating a clock pulse of frequency 2KHz on P1.2 by using Timer 0 Mode 1. Provide necessary calculation.	10

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Microprocessors And Microcontrollers [BECE204L] CAT-1 D1 2025-2026

Answer all questions					
Q. No.	Sub Sec.	Description	Marks	CO	BT Level
1		Which one do you prefer to design a low power, low cost smart irrigation system: Microprocessor or Microcontroller? Justify your selection with proper explanation.	[5]	1	K2
2		Explain the 8051 architecture in detail with a neat sketch.	[10]	3	K1
3		Consider the MPMC marks of 10 students are stored in internal RAM locations starting from 30H as given (20H, 15H, 25H, 33H, 45H, 50H, 39H, 25H, 28H, 49H). Write an 8051 assembly language program to find out the topper of the class and store it in an external memory location 3000H. The program should have the comments.	[10]	3	K3
4		In an 8051-based system, the serial communication is configured using Mode 1 (8-bit UART) with a baud rate of 9600 bps. a) What changes are required to double the baud rate to 19200 bps using the same 11.0592 MHz crystal? [2 Marks] b) Explain the role of SMOD bit in the PCON register during baud rate adjustment. [3 Marks]	[5]	4	K2
5		Write an 8051 assembly program that:	[20]	4	K3

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Microprocessors And Microcontrollers [BECE204L] CAT-1 B2 2024-2025

Q.No.	Sub. Sec.	Question	5	2
1.		Differentiate between a Microprocessor and a Microcontroller with respect to the following aspects: memory, resources, speed, architecture, and applications.	5	2
2.	(a)	With respect to 8086, explain the following signals. (i) $\overline{MN}/\overline{MX}$ (ii) $\overline{INTR}$ (iii) $\overline{DT}/R$ (iv) $\overline{TEST}$ (v) $\overline{BHE}$	5	
	(b)	Calculate the 20-bit physical address for each instruction VALUES = 3000H, CS = 0000H, DS = 2000H, SS = 3000H, ES = 6000H, IP = 8000H, SI = 4C00H, DI = 5D00H, SP = 5000H, BP = 2468H, AX = 2000H, BX = 3000H. (i) MOV AX, [246BH] (ii) MOV AX, [BX] [SI] (iii) MOV AX, 1234 [BX] [DI]	10	4
3.		An 8086 microprocessor is interfaced with an 8255 Programmable Peripheral Interface to control an LED and monitor two switches. The LED is connected to bit 2 of Port B, and the two switches are connected to bits 6 and 7 of Port A. The LED should turn ON when both switches are in the same state (either both HIGH or both LOW), and should remain OFF otherwise. Assume the address of the Port A is 0860H. Write an 8086 assembly language program to implement this logic, and draw a neat interfacing diagram showing the connections between the 8086, 8255, LED, and switches.	6	2
4.	(a)	Draw and explain the architecture of 8254.	4	
	(b)	Explain the memory segmentation of 8086 and respective registers for finding the 20-bit physical address.	10	4
5.	(a)	Write an 8086 assembly language program to generate the Fibonacci sequence. The length of the sequence is stored in the data segment, with the length specified as 10. Store the generated sequence in memory location starting from 4000H	5	
	(b)	Write an 8086 assembly language program to find the square root of perfect square number. Assume the input number is stored in the memory location 5000H, and the corresponding square root should be stored at 5001H.		

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Assume that 8051 internal RAM locations 30H – 34H have the following values, write an assembly program to find the sum of the values. At the end of the program, register A should contain the low byte and R7 the high byte of the sum.

30H = (7DH)
 31H = (EBH)
 32H = (C5H)
 33H = (5BH)
 10
 33

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
 CONTINUOUS ASSESSMENT TEST - I
 WINTER SEMESTER 2024-2025

Page 3 of 5
 SLOT: B1+TB1

KEY

34H = (30H)
 Key: Sum is A=B8H, R7=02H
 CODE:
 ORG 0000H
 MOV 30H,#7DH
 MOV 31h,#0EBH

```

MOV 32h,#0C5H
MOV 33H,#5BH
MOV 34H,#30H
MOV A,30H
ADD A,31H
JNC L1
INC R7
L1:ADD A,32H
JNC L2
INC R7
L2:ADD A,33H
JNC L3
INC R7
L3:ADD A,34H
JNC L4
INC R7
L4:SJMP L4
END

```

3. a) Indicate the addressing modes for the following 8051 instructions

- (i) MOV R0, #56H
- (ii) MOVC A, @A+DPTR
- (iii) MOV 38H, R0
- (iv) MOV R0, 38H
- (v) ADD A, @R1

Key:

- (i) MOV R0, #56H - IMMEDIATE ADDRESSING
- (ii) MOVC A, @A+DPTR - INDEXED INDIRECT
- (iii) MOV 38H, R0 - REGISTER ADDRESSING
- (iv) MOV R0, 38H - DIRECT ADDRESSING
- (v) ADD A, @R1 - INDIRECT ADDRESSING

5

3 3

b) Identify if the following 8051 instruction have any error, if so give the correct instructions

- (i) MOV @R2, A
- (ii) DEC DPTR
- (iii) CPL R3
- (iv) AND A, B
- (v) MUL A,B

5

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CONTINUOUS ASSESSMENT TEST - I

WINTER SEMESTER 2024-2025

Page 4 of 5

SLOT: B1+TB1

KEY

Key:

- (i) MOV @R2, A - MOV @R1, A
- (ii) DEC DPTR - INC DPTR
- (iii) CPL R3 - CPL A
- (iv) AND A, B - ANL A, B
- (v) MUL A,B - MUL AB

4. a) For the given 8051 assembly program, find out the time required to execute it, if XTAL frequency is 10 MHz.

MOV R4, #78H 1 machine cycle
 L1: ADD A, R4 1 machine cycle
 MOV 67H, 79H 1 machine cycle
 SUBB A, 67H 1 machine cycle
 INC A 1 machine cycle
 DJNZ R4, L1 2 machine cycle
 KEY: 721 MACHINE CYCLES IN DECIMAL
 L1 loop = (6x78H) = 02D0H = 720(DECIMAL)
 INCLUDING THE FIRST INSTRUCTION = 720 +1 = 721 (DECIMAL) IN HEXA IT
 IS 02D1H
 XTAL =10MHZ
 $f = 10\text{MHZ}/12 = 0.833 \times 10^6$
 $t = 1.2 \text{ microsecs}$
 total time = 1.2 microsecs x 721 = 865.54 microsecs

5

3 3

(b) Determine the value of A and B and status of the flag register after the execution of the following instructions:

ORG 0000H
 MOV A, #25H
 MOV B, #1FH
 MUL AB
 SUBB A, #0FCH
 RRC A
 END

KEY: A=BFH

B=04H

PSW = C1H [11000001] that is CY=1, AC=1, P=1, OV=0

5

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

CONTINUOUS ASSESSMENT TEST - I

WINTER SEMESTER 2024-2025

Page 5 of 5

SLOT: B1+TB1

KEY

5. (a) Examine the content of flags after performing logical OR operation of the following $(F1)_{16}$ OR $(1F)_{16}$

KEY: ALL FLAGS are 0

5

1 2

(b) Explain why microcontrollers are preferred over microprocessors for small to medium appliance applications.

Key: descriptive answer.

5

of 5

Analyze the following 8051 assembly program and show the stack and stack pointer during the execution of the program. Assume the default stack area.

MOV R6, #25H
 MOV R1, #12
 MOV R4, #0F3H
 PUSH 6



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Microprocessors And Microcontrollers [BECE204L] CAT-1 C



School of Electronics Engineering

Winter Semester 2023-24 Continuous Assessment Test - I

Programme Name & Branch: B.Tech (School of Computer Science Engineering)

SLOT: CI-TCI

Course Name & Code: Microprocessors and Microcontrollers BECE204L

Exam Duration: 90 Min.

Maximum Marks: 50

Q. No.	Questions	Max Marks
1	i) Nearly draw the block diagram of a microprocessor and briefly explain about all the blocks and their operations. ii) Identify and rectify (if any) the errors for the following instructions: DIV A,B MOV TIR, #155 MOV A, @R3 MOVC A, @A+PC MOV R7,R5	05
2	Write an 8051 assembly language program to transfer data from Rn where n=1, 2, 3 and 4 in register bank 1 to Rn where n= 3, 4, 5, and 6 in register bank 3 using stack operation.	10
3	Assume "your name" is stored in the code memory in hexadecimal number system. Write an 8051 assembly language program to transfer "your name" from code memory to data memory. Assume the source and destination address as required.	10
4	Write an 8051 assembly language program to get 8-bit number (x) from port P0. If x is odd number compute $y = x^2 + 5$ and if x is even number compute $y = x^2 + 2x$. Transfer the value y to port P1. Note: The value of x is such that y does not exceed 8-bits.	10
5	Write an 8051 assembly language program to create a waveform with 33% duty cycle on P0.2. Provide necessary calculations. Use Timer	10



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1. For 8051 Microcontroller, interpret the contents of various locations and the position of stack pointer after executing each and every instruction of the following code.

Line Number	Program	Contents
1	MOV R0H, #00H	00H 0F
2	MOV R1, #100H	00H 0F
3	MOV R2, #101H	00H 0F
4	MOV SP, #200H	0F 00
5	MOV R3, #00H	00H 0F
6	MOV R4, #00H	00H 0F
7	MOV R5, #00H	00H 0F
8	MOV R6, #00H	00H 0F
9	MOV R7, #00H	00H 0F
10	MOV R8, #00H	00H 0F
11	MOV R9, #00H	00H 0F
12	END	

2. Determine the size of delay in the following program, if the crystal frequency of 8051 microcontroller is 11.0592 MHz.

Label	Operands	Machine Cycle
L1	MOV R1, #00H	1
L2	MOV R2, #00H	1
L3	MOV R3, #00H	2
L4	MOV R4, #00H	2
L5	MOV R5, #00H	2

3. For 8051 Microcontroller, identify the error (if any) in the following instructions and discuss the error in detail.

- MOV R1,R2
- MOV R0,R1
- MOV R1,R2
- MOV R1,R2
- MOV R1,R2

4. Analyze the following code and specify the contents of various registers and Program Status Word (PSW) after the execution of each instruction.

Line	Instruction	PSW
1	ORG 00H	
2	MOV A, R0	A=00H, PSW=07
3	MOV R0, A	R0=00H
4	MOV R0, A	R0=00H
5	MOV R0, A	R0=00H
6	MOV R0, A	R0=00H
7	MOV R0, A	R0=00H
8	MOV R0, A	R0=00H
9	MOV R0, A	R0=00H
10	MOV R0, A	R0=00H
11	MOV R0, A	R0=00H
12	MOV R0, A	R0=00H
13	MOV R0, A	R0=00H
14	MOV R0, A	R0=00H
15	MOV R0, A	R0=00H
16	MOV R0, A	R0=00H
17	MOV R0, A	R0=00H
18	MOV R0, A	R0=00H
19	MOV R0, A	R0=00H
20	MOV R0, A	R0=00H
21	MOV R0, A	R0=00H
22	MOV R0, A	R0=00H
23	MOV R0, A	R0=00H
24	MOV R0, A	R0=00H
25	MOV R0, A	R0=00H
26	MOV R0, A	R0=00H
27	MOV R0, A	R0=00H
28	MOV R0, A	R0=00H
29	MOV R0, A	R0=00H
30	MOV R0, A	R0=00H
31	MOV R0, A	R0=00H
32	MOV R0, A	R0=00H
33	MOV R0, A	R0=00H
34	MOV R0, A	R0=00H
35	MOV R0, A	R0=00H
36	MOV R0, A	R0=00H
37	MOV R0, A	R0=00H
38	MOV R0, A	R0=00H
39	MOV R0, A	R0=00H
40	MOV R0, A	R0=00H
41	MOV R0, A	R0=00H
42	MOV R0, A	R0=00H
43	MOV R0, A	R0=00H
44	MOV R0, A	R0=00H
45	MOV R0, A	R0=00H
46	MOV R0, A	R0=00H
47	MOV R0, A	R0=00H
48	MOV R0, A	R0=00H
49	MOV R0, A	R0=00H
50	MOV R0, A	R0=00H

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Microprocessors And Microcontrollers [BECE204L] CAT-1 G2 2

General Instructions:																																								
Answer all the questions																																								
S. No	Question	Marks	CO	BL																																				
1. A	Draw the generalized architecture microcontroller and explain	5	1	1																																				
1. B	Write the functions of following registers and buses (i) Program counter (ii) Stack Pointer (iii) Data bus (iv) Address bus (v) Control bus	5	1	1																																				
2	Analyze the following 8051 assembly program which consist 16 lines and Comment for each line	10	3	2																																				
<table><tr><th>Line No.</th><th>Instruction</th><th>Line No.</th><th>Instruction</th></tr><tr><td>1</td><td>MOV R7, #25H</td><td>9</td><td>ADD A,B</td></tr><tr><td>2</td><td>SETB R5L</td><td>10</td><td>PUSH DTH</td></tr><tr><td>3</td><td>CLR R0</td><td>11</td><td>POP DTH</td></tr><tr><td>4</td><td>MOV R2,#12H</td><td>12</td><td>PUSH 17H</td></tr><tr><td>5</td><td>MOV A,#0EH</td><td>13</td><td>POP DTH</td></tr><tr><td>6</td><td>ADD A, R7</td><td>14</td><td>POP DCH</td></tr><tr><td>7</td><td>MOV DTH,#05H</td><td>15</td><td>POP DTH</td></tr><tr><td>8</td><td>MOV R,0FH</td><td>16</td><td>END</td></tr></table>					Line No.	Instruction	Line No.	Instruction	1	MOV R7, #25H	9	ADD A,B	2	SETB R5L	10	PUSH DTH	3	CLR R0	11	POP DTH	4	MOV R2,#12H	12	PUSH 17H	5	MOV A,#0EH	13	POP DTH	6	ADD A, R7	14	POP DCH	7	MOV DTH,#05H	15	POP DTH	8	MOV R,0FH	16	END
Line No.	Instruction	Line No.	Instruction																																					
1	MOV R7, #25H	9	ADD A,B																																					
2	SETB R5L	10	PUSH DTH																																					
3	CLR R0	11	POP DTH																																					
4	MOV R2,#12H	12	PUSH 17H																																					
5	MOV A,#0EH	13	POP DTH																																					
6	ADD A, R7	14	POP DCH																																					
7	MOV DTH,#05H	15	POP DTH																																					
8	MOV R,0FH	16	END																																					
3	Write an ALP to store the string "ALL IS WELL" from ROM address 200H to 25AH. Copy the string and store it in reverse order from RAM address 50H to 5AH.	10	3	3																																				
4	Assume a switch is connected at port pin P0.2, monitor the status of the switch. If the status of the switch is HIGH ("1") then transfer data from Port P1 to P3, else transfer data from P1 to R7 of Bank2.	10	4	3																																				

5	Write an 8051 assembly program to generate a signal at p1.5 with 4ms on time and time off time. Assume clock frequency of 8051 is 12 MHz and use the model timer T0.	10	4	3
---	--	----	---	---

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Analyze the following 8051 assembly program and show the stack and stack pointer during the execution of the program. The address of Bank0 = 00 to 07H, Bank1 = 08 to 0FH, Bank2 = 10 to 17H and Bank3 = 18 to 1FH. Assume the default stack area.

Answer:

Line

No

Instruction SP

Address

SP

Data

Line

No

Instruction SP

Address

SP

Data

1 SETB PSW.3 Bank 1 10 PUSH 1 1H 09H 0F3H

2 MOV R6, #25H 0BH 25H 11 PUSH 10H 0AH XXH

3 MOV R7, #12H 0FH 12H 12 PUSH 0FH 0BH 12H

4 SETB PSW.4 Bank 2 13 PUSH 0EH 0CH 25H

5 CLR PSW.3 14 POP 0DH 0DH 25H

6 MOV

R1, #0F3H

11H 0F3H 15 POP 0CH 0CH 12H

7 MOV

R2, #0DEH

12H 0DEH 16 POP 0BH 0BH XXH

8 MOV

R3, #0F7H

13H 0F7H 17 POP 0AH 0AH 0F3

9 PUSH 12H 0BH 0DEH 18 END


```
SJMP Begin
DELAY:CPL P1.5
SETB TR1
L1:JNB TF1,L1
CLR TF1
CLR TR1
RET
END
```

Note: We can use same ACALL DELAY for ON and OFF time of a signal because it is a square wave.

10

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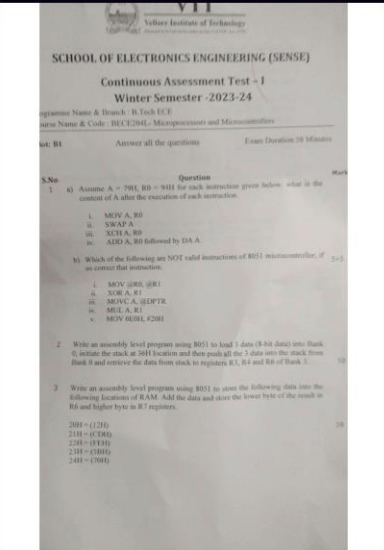
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Microprocessors And Microcontrollers [BECE204L] CAT-1 B1 2



SCHOOL OF ELECTRONICS ENGINEERING (SENSE)
Continuous Assessment Test - I
Winter Semester 2023-24
Programme Name & Branch: B.Tech ECE
Course Name & Code: BECE204L: Microprocessors and Microcontrollers
Set: B1 Answer all the questions Exam Duration: 30 Minutes

Q.No **Question** **Mark**

1 a) Assume $R_7 = 70H$, $R8 = 90H$ for each instruction given below, what is the content of A after the execution of each instruction.

i. MOV A, R8
ii. SWAP A
iii. XCH A, R8
iv. ADD A, R8 followed by DA A.

b) Which of the following are NOT valid instructions of 8051 microcontroller, if so correct that instruction.

i. MOV @R0, @R1
ii. XCH A, R1
iii. MOV A, @DPTR
iv. MOV A, R1
v. MOV @R0, #20H

2 Write an assembly level program using 8051 to load 3 data (8-bit data) into Bank 0, initiate the stack at 30H location and then push all the 3 data into the stack from Bank 0 and retrieve the data from stack to registers R3, R4 and R5 of Bank 1.

3 Write an assembly level program using 8051 to store the following data into the following locations of RAM. Add the data and store the lower byte of the result in R0 and higher byte in R7 registers.

200H = (123H)
210H = (456H)
220H = (789H)
230H = (ABCDEFH)
240H = (FEDCBAH)

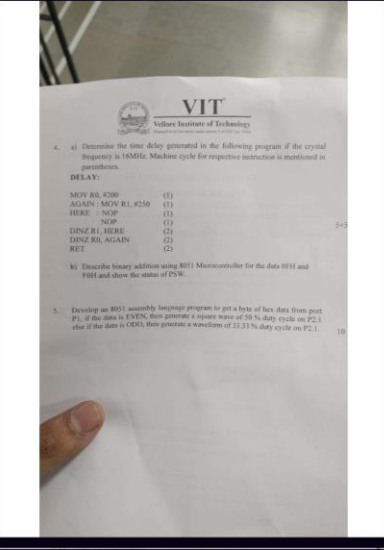
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Microprocessors And Microcontrollers [BECE204L] CAT-1 B1 2



VIT
Vellore Institute of Technology
Winter Semester 2023-24
Programme Name & Branch: B.Tech ECE
Course Name & Code: BECE204L: Microprocessors and Microcontrollers
Set: B1 Answer all the questions Exam Duration: 30 Minutes

4. a) Determine the time delay generated in the following program if the crystal frequency is 14MHz. Machine cycle for respective instruction is mentioned in parentheses.

MOV R0, #20H (1)
AGAIN: MOV R1, #25H (1)
HERE: NOP (1)
NOP (1)
DJNZ R1, HERE (2)
DJNZ R0, AGAIN (2)
RET (2)

b) Discuss the binary addition using 8051 Microcontroller for the data 0F1H and F0H and show the status of PSW.

5. Develop an 8051 assembly language program to get a byte of hex data from port P1, if the data is EVEN, then generate a square wave of 50% duty cycle on P2.1, else if the data is ODD, then generate a square wave of 33.33% duty cycle on P2.1.

[2 of 4](#) [100%](#) [🔍](#) [🔄](#) [📄](#) [🔍](#)

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S.No Question Marks CO BL

1 (a) Indicate the addressing modes for the following 8051 instructions

(i) MOV R0, #56H

(ii) MOVC A, @A+DPTR

(iii) MOV 38H, R0

(iv) MOV R0, 38H

(v) ADD A, @R1

(b) Identify if the following 8051 instruction have any error, if so give the correct instructions

(i) MOV @R2, A

(ii) DEC DPTR

(iii) CPL R3

(iv) AND A, B

(v) MUL A,B

5+5 CO3 L3

2 Analyze the following 8051 assembly program and show the stack and stack pointer during the execution of the program. Assume the default

stack area.

MOV R6, #25H

MOV R1, #12

MOV R4, #0F3H

PUSH 6

PUSH 1

PUSH 4

POP 7

POP 5

POP 3

END

10 CO3 L4

3 Assume that 8051 internal RAM locations 40H – 44H have the following values, write an assembly program to find the sum of the values. At the end of the program, register A should contain the low byte and R7 the high byte of the sum.

40H = (7DH)

41H = (EBH)

42H = (C5H)

43H = (5BH)

44H = (30H)

10 CO3 L6

4. (a) Write an 8051 assembly program to generate a waveform with 30% duty cycle on the port pin P1.2.

(b) For the given 8051 assembly program, find out the time required to execute it, if XTAL frequency is 11.0592 MHz.

MOV R4, #0FH 1 machine cycle

L1: ADD A, R4 1 machine cycle

MOV B, #09H 1 machine cycle

MUL A B 2 machine cycle

INC A 1 machine cycle

DJNZ R4, L1 2 machine cycle

5+5 CO4 L5

5. Assume a switch is connected at port pin P1.4, monitor the status of



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Microprocessors And Microcontrollers [BECE204L] CAT-1 G1 2024-2025


VIT
Vellore Institute of Technology
School of Electronics Engineering (SENSE)
Fall Semester 2024-25
Continuous Assessment Test - I
SLOT: G1+TG1

Programme Name & Branch: B. Tech & CSE (BCB, BCE, BCL, BCT, BDB, BKT)
Course Name & Code: Microprocessor & Microcontrollers (BECE204L)
Cen. Number: VL202420104021, VL202420104015, VL202420104023, VL202420104043, VL202420104013, VL202420104014, VL202420104034, VL202420104024, VL202420104025
Faculty Name (s): Debashish Das, A D D Dwivedi, G Raghav, Kalyanendra Ghosh, Vikas Vijayaraj, Bourdhan, Abhishek Tripathi, Nandini H. Lankar, Vishal Gupta
Exam Duration: 90 Min. Maximum Marks: 50

General Instructions:
Specify if any special material may be permitted
Any other specific instruction

Q. No.	Question	Max Marks
1.	Write an 8051 based program assuming the crystal frequency is 12MHz. Find the time required when if we want to have a time delay of 1ms? Also generate the time pulse width with equal amount of ON and OFF period using timer 1 and through its bit mode of operation.	10
2.	Write an 8051 assembly program to read port 0. If received data is equal to ANL, send the data "00H" to port 1, else send the data "55H" to port 1.	10
3.	a) Show the status of PSW register in 8051 after the subtraction performed between 03H and 04H. b) In a version of 8051, the crystal frequency is 11.0592MHz. Find the time delay associated with loop section of the following DELAY subprogram. DELAY: MOV R3, #200 HERE: NOP NOP NOP DJNZ R3, HERE RET	5+5
4.	Compare microprocessor with microcontroller with neat block diagram, explaining the blocks in detail.	10
5.	Write an 8051 assembly language program to add the odd numbers stored in code memory from the location 500H to 50FH. The stored data's are: 04H, 22H, 33H, A4H, C7H, 40H, 21H, 12H, 70H and 07H. Store the summation of odd numbers at location 02H and 03H of RAM.	10

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Identify, if the following 8051 instructions have any error. If so give the correct instructions:

(i) MUL A, B	5	3	2
(ii) PUSH A			
(iii) ANL R0, A			
(iv) MOV A, @R7			
(v) MOV R2, R1			
b). Identify the Addressing Mode of the following instructions:			
MOV 03H, 05H	5	3	2
MOV 03H, #05H			
MOV A, R7			
MOV A, @A+DPTR			
ADD A, @R0			

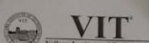
2. Write an assembly language program to add the following BCD numbers together and store the lower byte of the result in R3 and higher byte of the result in R4. Assume that the numbers are stored in the ROM space starting from 250H onwards.

ORG 250H
DATA: DB 54H, 76H, 51H, 98H, 88H, 99H

Notice that you must first bring the data from ROM space into the RAM and then add them together.

3. Develop an 8051 assembly language program to do the following task. Assume a switch is connected to port pin P1.0 and monitor its status continuously. If the status of the switch is high, transfer data from Port P0 to Port P2 and blink an LED connected to P1.1. If the switch status is low, transfer data from Port P0 to Port P3 and trigger a buzzer connected to P2.1. Use a delay sub-routine for blinking the LED or triggering the buzzer.

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4. Show the output of the following code after the execution and fill the results in Table 1

Table 1

Reg/Location	Value in Hex
R0	
R1	
R2	
R3	
R4	

5. (i) Analyse the following assembly code

```
SETB PSW.4  
MOV R2, #78H
```

10 3 3



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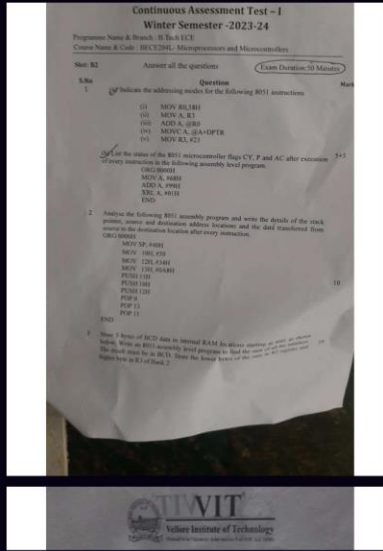
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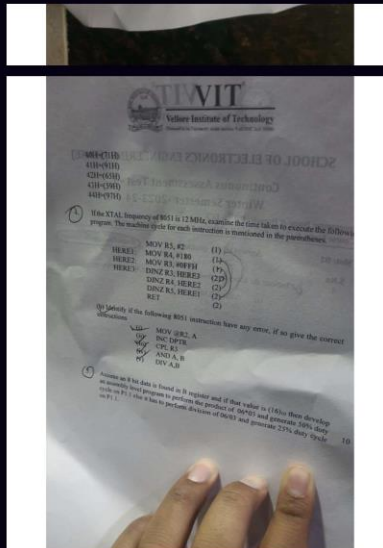
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